Time-Varying Covariates in Mortgage Survival: The Andersen–Gill Cox Extension

IR&C Assignment 3, Part E(ii) — Spring 2026

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1. Introduction

Standard Cox proportional hazards models fix all covariates at loan origination. This is inadequate for mortgage prepayment, where the key driver — the rate incentive $r_{\text{orig}} - r_{\text{market}}(t)$ — changes every month as market rates fluctuate. A borrower who locked in at 3% in 2020 had a strong refinancing incentive at origination; by 2023, as market rates rose to 7%, that same loan's incentive collapsed to -4 pp, making refinancing strictly dominated. A static model cannot represent this transition.

The **Andersen–Gill** (1982) extension resolves this by replacing one row per loan with one row per loan-month, updating covariates at each interval.

2. Data and Feature Engineering

Sample: 100,000 loans drawn from the monthly panel (stratified by vintage year), yielding 4,676,598 loan-month intervals (average ≈47 months per loan).

Features (13 total): nine static origination features (FICO, LTV, DTI, UPB, `orig_rate`, loan purpose dummies, occupancy dummies) shared by both models, plus four time-varying features updated monthly in TV Cox: `rate_incentive`, `unemployment`, `hpi_yoy`, and the synthetic current LTV (`ELTV`). The static Cox baseline uses the nine shared static features plus origination-era `unemployment` and `hpi_yoy` (11 features total). `mortgage_rate` is excluded from both models: since `rate_incentive` $= r_{\text{orig}} - r_{\text{market}}(t)$ and r_{orig} is fixed per loan, including both would create perfect within-loan collinearity.

Synthetic ELTV is computed monthly as:

$$\text{ELTV}_t = \text{LTV}_0 \times \underbrace{\frac{(1+r)^{360} - (1+r)^t}{(1+r)^{360} - 1}}_{\text{remaining balance fraction}} \div \underbrace{\prod_{s=1}^{t} \left(1 + \frac{\text{hpi\_yoy}_s}{1200}\right)}_{\text{cumulative HPI}}, \tag{1}$$

capturing equity build from scheduled amortisation and home-price appreciation — the two dominant drivers of the refinancing put option.

3. The Andersen–Gill Model

3.1 Hazard function

Each loan contributes one counting-process interval $(t_{\text{start}}, t_{\text{stop}}]$ per calendar month with the current covariate vector $\mathbf{x}_i(t)$:

$$h_i(t) = h_0(t)\, \exp\!\big(\boldsymbol{\beta}^\top \mathbf{x}_i(t)\big). \tag{2}$$

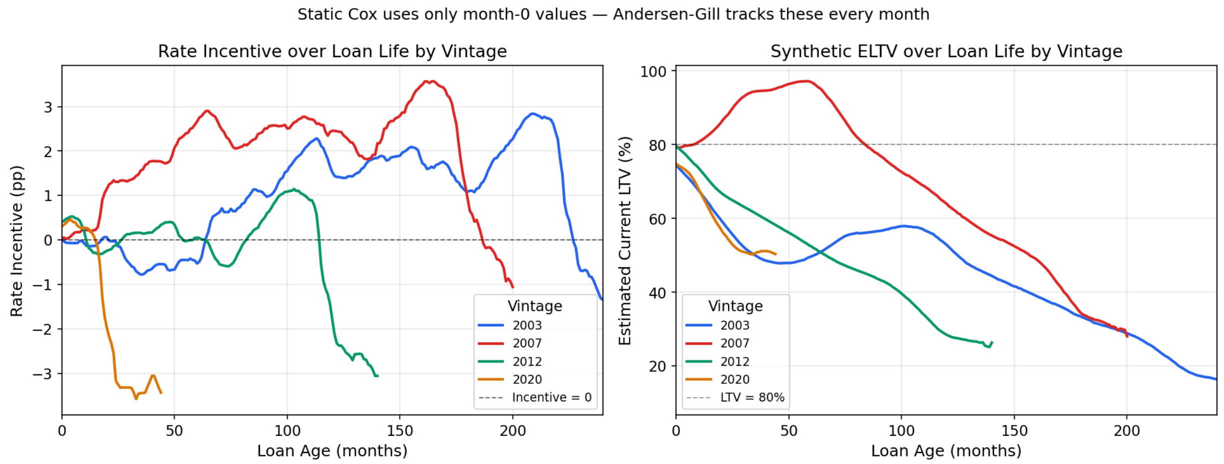

Figure 1: Median rate incentive (left) and ELTV (right) by vintage cohort over loan age. A static Cox model uses only month-0 values; the Andersen–Gill model tracks the full path. The 2020-vintage rate incentive turns negative within 3 years as market rates surged to 7%, while 2007-vintage ELTV rises steeply as home prices recover.

3.2 Partial likelihood

$$\ell(\boldsymbol{\beta}) = \sum_{i:\,\delta_i=1} \left[\boldsymbol{\beta}^\top \mathbf{x}_i(t_i) - \log \sum_{j \in \mathcal{R}(t_i)} \exp\!\big(\boldsymbol{\beta}^\top \mathbf{x}_j(t_i)\big) \right], \tag{3}$$

where $\mathcal{R}(t_i)$ is the set of all loan-months still active just before t_i. The key difference from the standard Cox partial likelihood is that each loan may contribute *multiple* intervals to the risk set, and $\mathbf{x}_i(t)$ is updated at each.

3.3 Landmark C-index

For time-varying models, the overall C-index conflates event timing with event probability. We instead compute the **landmark C-index** at fixed horizons t^*:

$$C(t^*) = P\Big(\hat{h}_i(t^*) > \hat{h}_j(t^*) \;\Big|\; T_i \le t^* < T_j\Big), \tag{4}$$

where each loan's score $\hat{h}_i(t^*) = \exp\!\big(\hat{\boldsymbol{\beta}}^\top \mathbf{x}_i(\min(T_i, t^*))\big)$ uses its covariate state at the landmark time. This aligns with Part B's per-horizon AUC evaluation.

4. Results

4.1 Coefficient comparison

Feature	Static Cox	TV Cox	Economic reading
`rate_incentive`	—	**Positive**	Primary refi channel; captures burnout and rate cycles invisible to origination snapshot
`unemployment`	Negative	**Positive**	Multicollinearity with Fed easing: high-unemp periods coincide with low rates
`hpi_yoy`	Positive	Positive, larger	Rising prices unlock equity prepayment
`ELTV`	—	Negative	Equity erosion suppresses refinancing

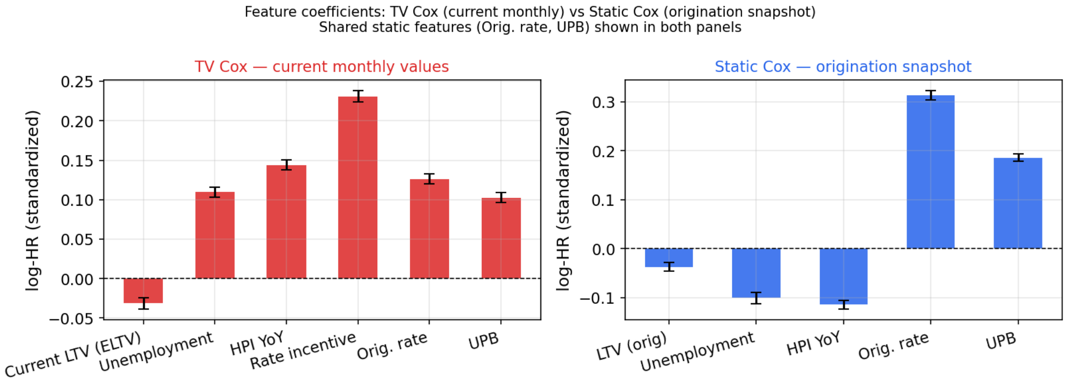

Figure 2: Coefficient comparison: TV Cox (left, red — 4 time-varying + `orig_rate` + UPB) vs. static Cox (right, blue — 2 origination-era macro + `orig_rate` + UPB). `orig_rate` and UPB are shared static features shown in both panels. Both macro signals flip sign from static to TV Cox — high-unemployment origination cohorts pre-date Fed easing, while current high-unemployment months coincide with low rates.

4.2 Model performance

Model	Log-L	C@T=12	C@T=24	C@T=36	C@T=60	C@T=84	C@T
Static Cox (origination snapshot)	−676,547	0.609	0.627	0.630	0.625	**0.623**	
TV Cox — Andersen–Gill	−674,551	**0.681**	**0.687**	**0.675**	**0.646**	0.605	
Part B Macro Cox (1.85M loans, overall C)		0.6428 (not per-horizon; shown for scale only)					

ΔLog-L = +1,996 confirms that monthly-updated covariates add genuine information beyond the origination snapshot. TV Cox leads at $T \leq 60$; static Cox overtakes at $T = 84$ (-0.018) and $T = 120$ (-0.037), with the crossover near $T \approx 70$ months.

5. Key Findings and Insights

1. **Time-varying covariates add real information.** ΔLog-L = +1,996 (TV vs. static, 100K loans) and a +0.060 C-index gain at $T = 24$ confirm that monthly-updated features capture variation invisible to the origination snapshot.

2. **Short- and medium-horizon gains are large; static Cox overtakes beyond $T \approx 70$ months.** TV Cox leads by +0.072 at $T = 12$, +0.060 at $T = 24$, +0.045 at $T = 36$, and +0.021 at $T = 60$, then falls below static at $T = 84$ (-0.018) and $T = 120$ (-0.037). Static Cox is remarkably stable (0.609–0.630 across all horizons), while TV Cox degrades monotonically past the 2-year refi peak.

3. **Long-horizon degradation is empirically confirmed, not just theoretical.** At $T = 84$ and $T = 120$, surviving loans are a selected group of locked-in, high-ELTV borrowers whose current covariate state reflects *why they have not prepaid*, not their future risk. The landmark super-model — conditioning on survival to a reference age s then predicting over $(s, s + \tau]$ — is the standard fix.

4. **Rate incentive is the key time-varying channel.** The current spread $r_{\text{orig}} - r_{\text{market}}(t)$ captures burnout and rate cycles invisible to the origination snapshot. As the primary rate variable in TV Cox, its coefficient measures the marginal refi incentive directly.

5. **Unemployment sign flip warns of multicollinearity.** High-unemployment periods (2009–2012) coincided with Fed easing and low mortgage rates, so unemployment partially proxies for loose monetary policy. Always inspect TV coefficients against macroeconomic narratives, not just sign.

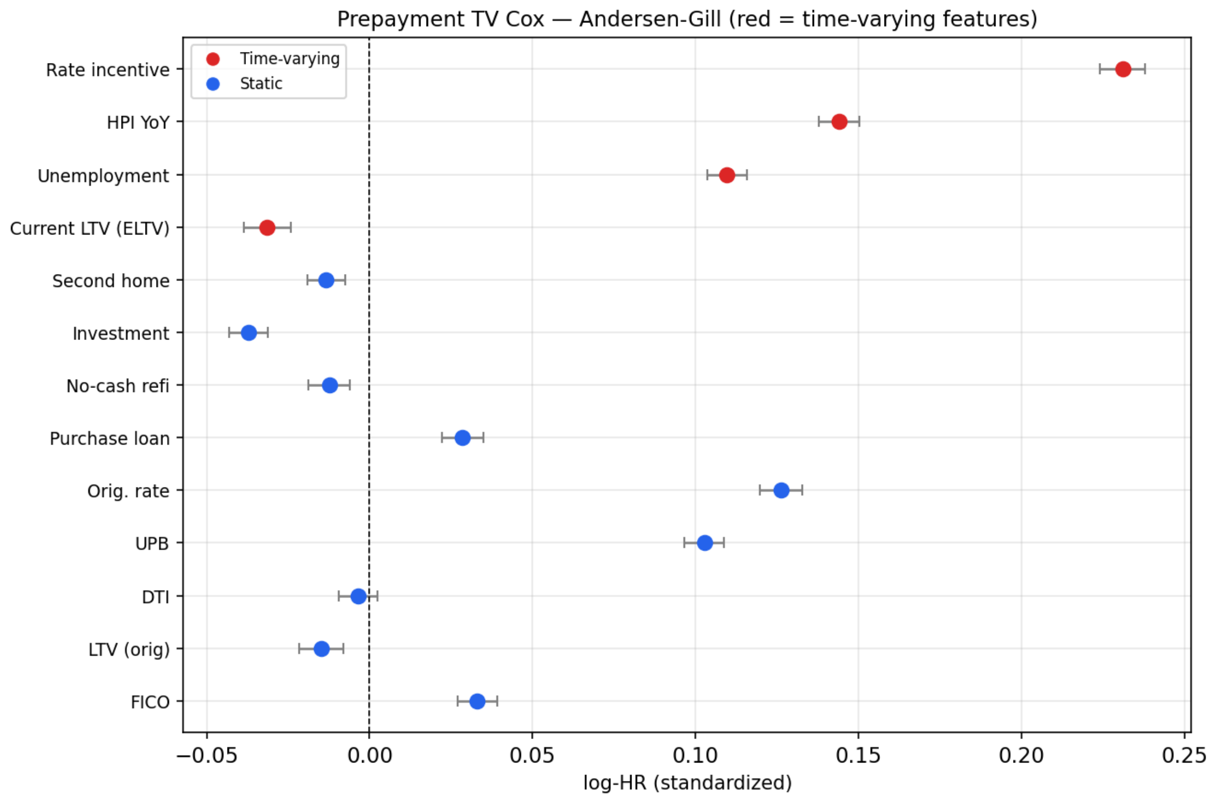

Figure 3: TV Cox forest plot. Red markers: time-varying features; blue markers: static origination features. `rate_incentive` is the dominant positive predictor of prepayment; `ELTV` is negative (high current LTV suppresses refinancing).

Prepayment: Static Cox vs Andersen-Gill TV Cox [50K loans · landmark C-index at fixed horizons]

Model	Log partial-L	C-index T=12	C-index T=24	C-index T=36	C-index T=60	C-index T=84	C-index T=120
Static Cox (origin.)	-676,547	0.6088	0.6273	0.6301	0.6252	0.6226	0.6203
TV Cox — Andersen-Gill	**-674,551**	**0.6812**	**0.6868**	**0.6753**	**0.6458**	**0.6046**	**0.5825**

Figure 4: Landmark C-index at fixed horizons: Static Cox vs. TV Cox Andersen–Gill (100K loans, 10K-loan evaluation subsample). TV Cox leads at $T = 12$–60 (peak $+0.072$ at $T = 12$, $+0.060$ at $T = 24$); static Cox overtakes at $T = 84$ and $T = 120$ as the landmark contamination effect overwhelms the TV signal for long-surviving loans.

6. **Synthetic ELTV adds meaningful signal.** Equity-unlocking events from HPI appreciation trigger cash-out refis invisible to static origination LTV. The negative ELTV coefficient in TV Cox confirms this channel.

Priority improvements: (1) Landmark super-model conditioning on survival to a reference age — empirically required given the $T = 84/T = 120$ degradation observed here. (2) Vintage-based out-of-time train/test split to produce an unbiased benchmark. (3) Rolling delinquency indicator — the largest single missing signal from E(i) ($|\Delta\hat{\beta}| = 0.543$).

References

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[2] Sadhwani, A., Giesecke, K. and Sirignano, J. (2021). *J. Financial Econometrics* 19:313–368.

[3] Cox, D. R. (1972). Regression models and life tables. *JRSS-B* 34:187–220.